

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Metallurgical and Materials Engineering
2. Subject Name: Transport Phenomena in Metallurgical Processes
L-T-P: 3-0-0 Credits: 3
3. Pre-requisites: TRANSPORT PHENOMENA IN METALLURGICAL PROCESSES
4. Syllabus and reference books:

Syllabus:

Momentum Transfer in Fluid Flow

Fluids and fluid flow characteristics, Estimation of viscosity of gases, gas mixtures, molten metals and slags.

Differential mass and momentum balance: Equation of continuity and Navier Stokes' equation, application for few simple cases.

Turbulent Flow: Concept of friction factor, Velocity boundary layer, Flow through packed and fluidized beds.

Overall energy balance (Bernoulli's equation): Energy conservation and friction losses, flow meters, flow from ladles and supersonic nozzle.

Heat Transfer

Steady and unsteady heat conduction equations and their applications in composite walls, finite, semi-infinite solid bodies.

Heat transfer with change of phases (solidification), Convective heat transfer and general heat balance equation, Thermal boundary layer, Heat transfer coefficient and dimensionless correlations.

Laws of radiative heat transfer, view factor, equivalent circuit and heat exchange between black and grey surfaces of different orientation, combined radiative and conductive heat transfer.

Mass Transfer

Concept of mass diffusion, factors affecting diffusivity in solids, liquid metals and gases, diffusion through solid and porous materials

Reference Books:

- 1) Transport Phenomena in Materials Processing, D. R. Poirier and G. H. Geiger, 2016, TMS (Springer).
- 2) Introductory Transport Phenomena, R. B. Bird, W. E. Stewart, E. N. Lightfoot and D. J. Klingenberg, 2015, Wiley.
- 3) Transport and Chemical Rate Phenomena, Nickolas J. Themelis, 1995, Gordon and Breach Publishers.

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Transport Phenomena and Rate Processes, Study scales of Transport Phenomena, Tools used to study Transport Phenomena, Diffusive, Convective and Radiative Flux, Phenomenological Laws and their Analogous Forms	3
2.	Fluids Properties and Types of Fluid Flow, Flow Field, Estimation of Viscosity of Gases by Kinetic Theory, Viscosity Estimation using Intermolecular Force Parameters and Critical Parameters, Viscosity of molten metals & alloys, slags, silicates and salts.	4
3.	Control Volume System and Deformation and Rotation Rate, Microscopic Balance: Mass Conservation (Continuity Equation) and Momentum Conservation (Navier Stokes' Equation, NSE), NSE Applications in Simple Laminar cases: Flows in Parallel Plates, Pipe and Falling Film, Stream Function and Velocity Potential, Stokes' law of drag force and its applications, Velocity Boundary Layers.	4
4.	Turbulent Flow: Equation of motion for turbulent flow, Friction Factor for Flow in Tubes Dimensional analysis for friction factor, Flow past a sphere and other submerged objects, Flow through Porous Media, Darcy's Law, Tube-bundle theory and Ergun's equation, Correlation in Fluidized beds	4
5.	Macroscopic Energy Balance in Fluid Flow, Frictional energy loss in straight conduits (Fanning Equation), Valves and Fittings (Fixture), Flow Measurement: Velocity meters, Head meters, Area meters, Fluid Flow from Ladles, Supersonic Nozzles and Jet Behavior	4
6.	Heat Transfer by Conduction: Energy Equation for Conduction, Steady-State 1-Dimensional Heat Conduction in Simple Cases, The Plane Wall without Generation of Heat, Concept of heat transfer co-efficient, multilayered wall with convection at the boundary surfaces, Radial Heat transfer through Hollow Infinite Cylinder, Composite cylindrical wall, Critical Thickness of Insulation	4
7.	1-Dimensional Unsteady Heat Conduction: Newtonian heating/cooling, Cooling of Infinite solids with Internal Temperature Gradient: Series and graphical solution of infinite, Transient Heat Transfer from Surface: Semi-infinite solid, Transient Temperature in Simple Finite Multidimensional Solids	4
8.	Convective Heat Transfer: General Energy Equation, Thermal Boundary Layers: Laminar forced and natural convection, Rayleigh's Method: Dimensional analysis of forced and natural convective heat transfer coefficients	3
9.	Heat transfer with change of phases (solidification): Solidification in Sand Molds, Metal Molds, Continuous Casting, Crystal Growth.	3
10.	Radiation Heat Transfer: Black body radiation, Absorptivity and Emissivity, Radiation Intensity and Emissive Power, view factor, equivalent circuit and heat exchange between black and grey surfaces of different orientation, combined radiative and conductive heat transfer.	3
11.	Mass Transfer: Concept of mass diffusion, factors affecting diffusivity in solids, liquid metals and gases, diffusion through solid and porous materials	2
Total number of hours		38